

Ch1 投资与估值概述

Introduction to Investing and Valuation

第一章 投资与估值概述

stock valuation / financial statements user: investor, manager, government, supplier

Investment Styles

Fundamental Investing: Challenge market prices

基本面投资

- *Active investing* : to find mispriced investments
- *Defensive investing* : a defense against paying the wrong price (or selling at the wrong price)

市场中证券存在**错误定价mispriced**, 分为 **定价过高overpriced** 和 **定价过低underpriced**

因此 券商会 对证券进行评级, 给出一个valuation, 从而与market price进行比较, 来进行判断买入、卖出或持有

贝塔风险：基本面风险

- 可用资本资产定价模型CAPM计算出, 代表**平均收益**
 - Required return = risk-free return + premium for return
 - Premium for return = risk premium on risk factors * sensitivity to risk factors
- 多元化投资能够消除基本面风险, 而且能通过多元化投资消除掉的这部分风险, 市场是不会给予回报的。
- passive investor: β

阿尔法风险：价格风险

- 对市场定价的合理性进行检验, 挖掘**超额收益** (比beta风险对应的收益水平更高的期望报酬的股票)
- active investor: $\beta + \alpha$

指数投资：被动投资的一种极端形式

- 购买股票市场组合 类似于标普500 上证50 沪深300
- 特点：市场组合极大地分散了投资

Bubble, Toil and Trouble

泡沫、陷阱和麻烦

How bubble works

连锁信游戏 - 每一个加入连锁信游戏的人都需要向他的上家支付一定的金额，这样的计谋就被称为庞氏骗局(Ponzi scheme).

不可信分析

- 不重视利润
 - 利润和商业模式一样重要
- 分析人员称一家公司价值主要在“无形资产”，却说不出怎样能估算出这些无形资产的价值
 - 知识确实是有价值的，主要体现在知识能生产出产品和服务，而产品和服务能带来销售收入，最终通过销售收入能创造利润
- 过度依赖一些非财务计量指标
 - 网页浏览量、使用量、客户数、设备使用率
 - 虽然能暗示获利能力，但不是获利的保证
 - 分析人员有责任指出这些指标如何转化成未来的盈利
- 不重视历史数据
- 只看局部，不看整体
- 不重视企业利润的质量，忽视盈利数据的会计质量问题

The Setting: investors, firms, securities and capital markets

环境：投资者、企业、证券与资本市场

企业价值

企业价值 = 负债价值 + 股权价值

$$V_0^F = V_0^D + V_0^E$$

- 资产 = 负债 + 所有者权益

- 负债的估值一般都比较准确

企业创造价值的三种主要活动

- 融资活动 financial activities
- 投资活动 investing activities
- 经营活动 operating activities

The Analysis of Business

Opinion: the valuation for bond is more difficult than predicting a price for stock.

兼并公司：买入很多项目群

诚讯金融 许国庆 金融估值建模培训

铂涛集团 连锁酒店 - 对不同消费者开发不同酒店品牌 针对不同需求 **市场分割**

评估前提

- Understand the business model(strategy) - know the business: know the firm's products
 - Product: 种类、消费者需求、价格弹性(是否有价格权)、产品的替代性、产品品牌名称关联、产品专利保护
 - Technology: BYD
 - knowledge base:
 - industry competition: HHI行业的集中程度 C4行业中前四名的市场份额占比 L8
 - management
 - Political, legal and regulatory environment
 - Sustainability of competitive advantage
- master the details

Financial statements are the lens on the business. *财务报表是商业的透镜。*

估值：基于基本面分析得知的战略、企业的财务报表，使用恰当的估值技术

估值核心：如何计算超额价值extra value

估值技术的选取要求：根据 成本-效益 原则

常用的估值技术

不涉及预测的估值技术

- 乘数比较法 CH3
 - 通过能观察到的可比公司价格乘数(P/E P/B P/S)来确定一家企业的股票价值
 - P/E 市净率 P/B 市盈率 P/S 市销率
- 乘数筛选法 CH3
 - 根据相关乘数判断股价的高低
 - for instance, 买入市盈率P/B低的, 卖出P/B高的
- 以资产为基础的估值 CH3
 1. 估算企业各项资产的公允价值, 加总
 2. 再减负债的价值

涉及预测的估值技术

- 股利贴现模型: 预测股利 CH4
 - 股票价值 = 未来股利现值
- 贴现现金流量分析: 预测自由现金流量 CH4
 - 股票价值 = 未来自由现金流量现值
- 剩余收益分析: 预测盈利和账面价值 CH5
 - 股票价值 = 当前权益账面价值 + 未来剩余收益的现值
- 盈利增长分析: 预测盈利和盈利的增长 CH6
 - 股票价值 = 资本化的盈利 + 未来超常增长收益的现值

Ch2 财务报表简介

Introduction to the Financial Statements

财务报表简介

The form of financial statements

- **balance sheet** 资产负债表
 - 反映存量 stock
 - $\text{Assets} = \text{liabilities} + \text{shareholder's equity} \rightarrow \text{Value of firm} = \text{value of debt} + \text{value of equity}$
- **income statement** 利润表
 - 反映流量 flows
 - $\text{Revenues} - \text{cost of goods sold} = \text{gross margin (毛利润)}$
 $\text{gross margin} - \text{operating expenses} = \text{operating income (EBIT)}$
- **cash flow statement** 现金流量表(资金来源三方面)
 - $\text{change in cash} = \text{cash from operations 运营} + \text{cash from investment 投资} + \text{cash from financing 融资}$
- **statement of stockholder's equity** 股东权益变动表(ending-beginning)
 - $\text{Ending equity} = \text{beginning equity} + \text{comprehensive income} - \text{net payout to shareholders}$
 - **Comprehensive income = net income + other comprehensive income**
 $\rightarrow$ dirty surplus accounting
 - $\text{net payout to shareholders} = \text{cash dividends} + \text{share repurchases} - \text{share issues}$

巨潮资讯 - 所有上市公司的年报 对比国内上市公司财务报表与课本内美国公司的

Measurement in the financial statements

Price-to-Book Ratio 市净率

$\text{value of equity} = \text{value of the firm} - \text{value of debt}$

$\text{intrinsic premium 内在溢价} = \text{intrinsic value of equity} - \text{book value of equity}$

Market premium 市场溢价 = market price of equity - book value of equity

- 若premium<0, 则称为discount折价
- 有人收购该企业时, 支付的premium溢价体现在资产负债表中的一项资产——商誉 goodwill

P/B = market price / book value

- 中位数P/B>1, 说明存在溢价。

Price-to-Earnings Ratio 市盈率

P/E = market price / earnings

- price代表未来盈利能力, earnings代表当前盈利能力, 比值代表对未来盈利增长的预期

Measurement in the Balance Sheet

公允价值(fair value)计量

- 盯市会计 - 与市场价格有关

历史成本(historical cost)计量

- 不会或不能按市价来报告它们的价值

premium产生原因

- 资产负债表中净资产无法体现企业所有价值 (漏了很多价值)
- 因此可以从未来的现金流量去计算

Measurement in the Income Sheet

revenue recognition principle 收入确认原则

- 只有把东西卖出去了 (确认了收入), 才视为价值的增加

matching principle 配比原则

- 将费用与收入进行配比, 来计算价值增加净额

Accounting as an anchor: Don't mix what you know with speculation

value = anchoring accounting value + speculative value

- anchoring accounting value = minimum true value

- Coming from financial statements / accounting reports / ...

Ch3 财务报表在估值中的应用

How Financial Statements Are Used in Valuation

财务报表在估值中的应用

Multiple Analysis 乘数分析

最简单的估值技术：使用的信息最少

常用乘数(multiples)

- 一般是 the stock price 与 earnings / book values of equity / sales / cash flow from operations 的比值

下面是以乘数为依据的两种乘数分析技术

Method of Comparables 乘数比较法

Steps

1. 选取 **Target Firm**，再根据 similar operations 选取 **Comparable Firms**
2. 计算 comparable firms 的一些 **mutiples**
3. 取multiples的平均值或中位数，再与 target firm 的相应指标相乘，得到估值

Screening on Multiples 乘数筛选法

Steps

1. 对 **screen stocks** 选取一个 multiple
2. 将 **multiple** 从高到低排序
3. 卖出高的，买入低的

Asset-Based Valuation 以资产为基础的估值

Essential: value of equity = value of firm - value of debt

- Value of firm = add up all the assets
- But there are some intangible assets: brand assets/ knowledge assets/ managerial assets -- 0 in balance sheet

 以上的估值方法共同点：都不涉及forecast

Fundamental Analysis 基本面分析

基本面分析五部曲 (process)

1. Knowing the business $\Rightarrow$ **strategy** to add value
2. Analyzing information
3. Developing forecasts
4. Converting the forecast to a valuation
 - 需要用到 **discount rate/ required return/ cost of capital** 资本成本率 -- 将资金用于某个项目的机会成本率
 - 这三个是同一个东西，看问题的角度不一样而已
5. Trading on the valuation
 - Outside investor
 - Compare value with price to buy, sell, or hold
 - Inside investor
 - Compare value with cost to accept or reject strategy

The Valuation Model -- The Architecture of Fundamental Analysis

Terminal investments and going-concern investments

Terminal: bond valuation/ project valuation

$$V_0^D = \frac{CF_1}{\rho_D} + \frac{CF_2}{\rho_D^2} + \frac{CF_3}{\rho_D^3} + \frac{CF_4}{\rho_D^4}$$

quorum. $D : Debt$; $\rho_D = 1 + RequiredReturn$

$$V_0^P = \frac{CF_1}{\rho_P} + \frac{CF_2}{\rho_P^2} + \frac{CF_3}{\rho_P^3} + \frac{CF_4}{\rho_P^4}$$

quorum. $P : Project$; $\rho_D = 1 + RequiredReturn$

Going-concern: stock valuation

Valuation Models, the Required Return, and Asset Pricing Model

Required return = risk-free rate + risk premium

Required return = risk-free rate + [Beta * market risk premium]

Criteria for a Practical Valuation Model

- Finite forecast horizons 有限预测期
- Validation 可证实性
 - 用到的东西一定要能够在财务报表上得到证实
- Parsimony 操作简洁

Value Generation

- Operating Activities
- Investing Activities
- some Financing Activities
 - Equity Financing Activities
 - share issues in efficient market ⚠ efficient market: 股票的市场价格能够反映它们的价值
 1. 原价发行 -- 每股价值不变
 2. 折价发行 -- 每股价值下降（如管理层向员工发行股份）
 - share issues in inefficient market
 1. 发行股票 -- 往往是在价值被市场高估时 -- 原股东赚，但会被市场当作坏消息而引起股价下跌
 2. 股票回购 -- 往往是在价值被市场低估时 -- 从而引起股价上涨
 3. 发放股利 -- 并不创造价值 -- 但存在 dividend signaling -- 股利多会被当作利好
 - return to shareholders = dividends + capital gains --
dividend⬆ capital gain⬇
 - Debt Financing Activities
 - debt financing is **irrelevant** to the value of the firm because selling debt is to raise money in a fair value.
 - Controversy: tax shield

Formula of Ch5

1. Value

Value = Book value + Premium

2. Residual earning (RE)

剩余收益 Residual earning a.k.a. 剩余盈利 Residual income

$$\text{Residual earning}_1 = \text{Earning}_1 - (\text{Required return} \times \text{Investment}_0)$$

Residual earnings model 剩余收益模型

$$\text{Value} = \text{Book value} + PV(\text{Expected RE})$$

3. Value of common equity (V_0^E)

$$V_0^E = B_0 + \frac{RE_1}{\rho_E} + \frac{RE_2}{\rho_E^2} + \frac{RE_3}{\rho_E^3} + \dots + PV(CV_{T+1})$$

1. where CV is continuous value (持续价值)

a. $RE_T = 0$, $CV_{T+1} = 0$, 溢价水平 $V_T^E - B_T = 0$

b. $RE_T \neq 0$, 且稳定, $CV_{T+1} = \frac{RE_{T+1}}{\rho_E - 1}$, 溢价水平 $V_T^E - B_T = \frac{RE_{T+1}}{\rho_E - 1}$

c. $RE_T \neq 0$, 不稳定, $CV_{T+1} = \frac{RE_{T+1}}{\rho_E - g}$, $g=1+\text{growth rate of RE}$

2. where RE is residual earnings = Comprehensive earnings - (Required return for equity * Beginning-of-period book value)

$$RE_t = \text{Earnings}_t - (\rho_E - 1)B_{t-1}$$

ρ_E =discount factor (贴现因子)

As $RE > 0$, the shares of this firm are worth a premium over book value(股票价值>账面价值).

3. Valuation: **Perpetuity:** $D_0 = 100\%$, $V_0^E = B_0 + \frac{RE_1}{\rho_E - g} (/ \rho_E)$

where D is dividend, $D_1 = D_0 + \text{growth rate of RE}$, ($D_2 = D_1^2$, $D_3 = D_1^3, \dots$)

At this time (perpetuity), growth rate of RE MUST be constant.

4. ROCE (rate of return on common equity)

Residual earnings = (ROCE - Required return on equity) * Book value of common equity

$$RE_t = Earn_t - (\rho_E - 1)B_{t-1} = [ROCE_t - (\rho_E - 1)]B_{t-1}$$

where $ROCE_t = \frac{Earn_t}{B_{t-1}}$ (综合收益/账面价值)

从后边的学习我们可以知道: $B=CSE$

- If *Expected RE* > 0 (*ROCE* > *cost of capital*), must at a premium. (溢价) C5.1.
- $\text{cost of capital} = \rho_E - 1$

5. P/B ratio (市净率)

市净率=每股市价/每股净资产

$$P/B \text{ ratio} = \frac{\text{Price}}{\text{Book value}}$$

- $P/B=1 \Rightarrow \text{Price per share} = \text{Book value} \Rightarrow \text{Premium(溢价)}=0 \Rightarrow ROCE=\rho_E - 1$ (cost of capital)

6. per common share

D

i.e.

$$\Delta \frac{\text{Book value}}{\text{Share}} = \frac{\text{Earnings}}{\text{Share}} - \frac{\text{Dividend}}{\text{Share}}$$

Thus,

$$ROCE_t = \frac{EPS_t}{BPS_{t-1}}$$

$$RE_t = [ROCE_t - (\rho_E - 1)]BPS_{t-1} = EPS_t - (\rho_E - 1)BPS_{t-1}$$

- 所有ROCE用Earning算更加准确（减小误差）

Formula of Ch6

1. Prototype valuation

1. 资本化处理（流量概念→存量概念）

$$\text{Capitalized earning} = \frac{\text{Earnings}}{\text{Required return}}$$

2. Valuation based on earnings

$$\text{Value} = \text{资本化盈利} + \text{预计盈利增长能带来的额外价值}$$

2. Value a share

超常收益增长 = 含息收益 - 正常收益

D

where

D

D

where $(\rho_E - 1)\text{dividend}_{t-1}$ is 上一年度再投资股息的收益

Principle:

If growth rate of Cum-div E = $\rho_E - 1$, asset's value=capitalized earnings

⇒

Value of equity = Capitalized forward earnings + Extra value for abnormal Cum-div E growth

$$\begin{aligned} V_0^E &= \frac{\text{Earn}_1}{\rho_E - 1} + \frac{1}{\rho_E - 1} \left[\frac{\text{AEG}_2}{\rho_E} + \frac{\text{AEG}_3}{\rho_E^2} + \frac{\text{AEG}_4}{\rho_E^3} + \dots \right] \\ &= \frac{1}{\rho_E - 1} \left[\text{Earn}_1 + \frac{\text{AEG}_2}{\rho_E} + \frac{\text{AEG}_3}{\rho_E^2} + \frac{\text{AEG}_4}{\rho_E^3} + \dots \right] \end{aligned}$$

STEP:

step1. Forecast earnings of Year 1.

step2. Forecast AEG of Year 2+.

step3. Calculate the PV(expected AEG).

step4. Capitalized earning = sum of AEG(/required return)

⇒

$$\begin{aligned}\text{Abnormal earnings growth}_t &= \text{Cum-dividend earnings}_t - \text{Normal earnings}_t \\ &= [\text{Earn}_t + (\rho_E - 1)\text{dividend}_{t-1}] - \rho_E \text{Earn}_{t-1}\end{aligned}$$

3. AEG model

1. If no AEG is forecasted,

$$V_0^E = \frac{\text{Earn}_1}{\rho_E - 1}$$

thus,

$$P/E = \frac{V_0^E}{\text{Earn}_1} = \frac{1}{\rho_E - 1}$$

2. If AEG is forecasted, when $AEG > 0$, $P/E > \frac{1}{\rho_E - 1}$.

4. P/E (normal)

Price = Value at end = Value at beginning + Earnings – Dividends

when $AEG=0$ or 溢价=0,

1. Trailing P/E 既往市盈率:

$$\text{Trailing } P/E = \frac{\text{Price} + \text{dividend}}{\text{Earnings}} = \frac{\text{Value}_{-1} \times [1 + (\rho_E - 1)]}{\text{Value}_{-1} \times (\rho_E - 1)} = \frac{\rho_E}{\rho_E - 1}$$

2. Forward P/E 远期市盈率:

$$\text{Forward } P/E = \frac{\text{Price}}{\text{Forward earnings}} = \frac{\text{Price}}{\text{Price} \times (\rho_E - 1)} = \frac{1}{\rho_E - 1}$$

5. A poor P/E model

$$V_0^E = \frac{\text{Earn}_1}{\rho_E - g}, \quad g=1+\text{growth rate of Earning}$$

With this model,

$$\text{Forward } P/E = \frac{1}{\rho_E - g}$$

6. PEG ratio

P/E-to-earnings-growth 市盈率/预计下一年度的收益增长率

$$\text{PEG ratio} = \frac{P/E}{\text{1-year-ahead percentage earnings growth}}$$

Formula of Ch8

Cash conservation equation 现金恒等式

Free cash flow = Net dividends to shareholders + Net payments to debtholders and insurers

$$C - I \equiv d + F$$

where

C = Net cash from operations 经营产生现金流

I = Cash investment 现金投资额

$C - I$ = Free cash flow 自由现金流

d = Equity financing flow 权益融资现金流

= Net dividends = Cash dividends + Share repurchases – Share issues

F = Debt financing flows 债务融资现金流（流出pay为正，流入receive为负）

= Net cash outflow for debtholders & debt insurers

= Net principal payments

Treasurer's rule 财务主管法则

Cash applied to debt trading = $(C - I - i) - d$

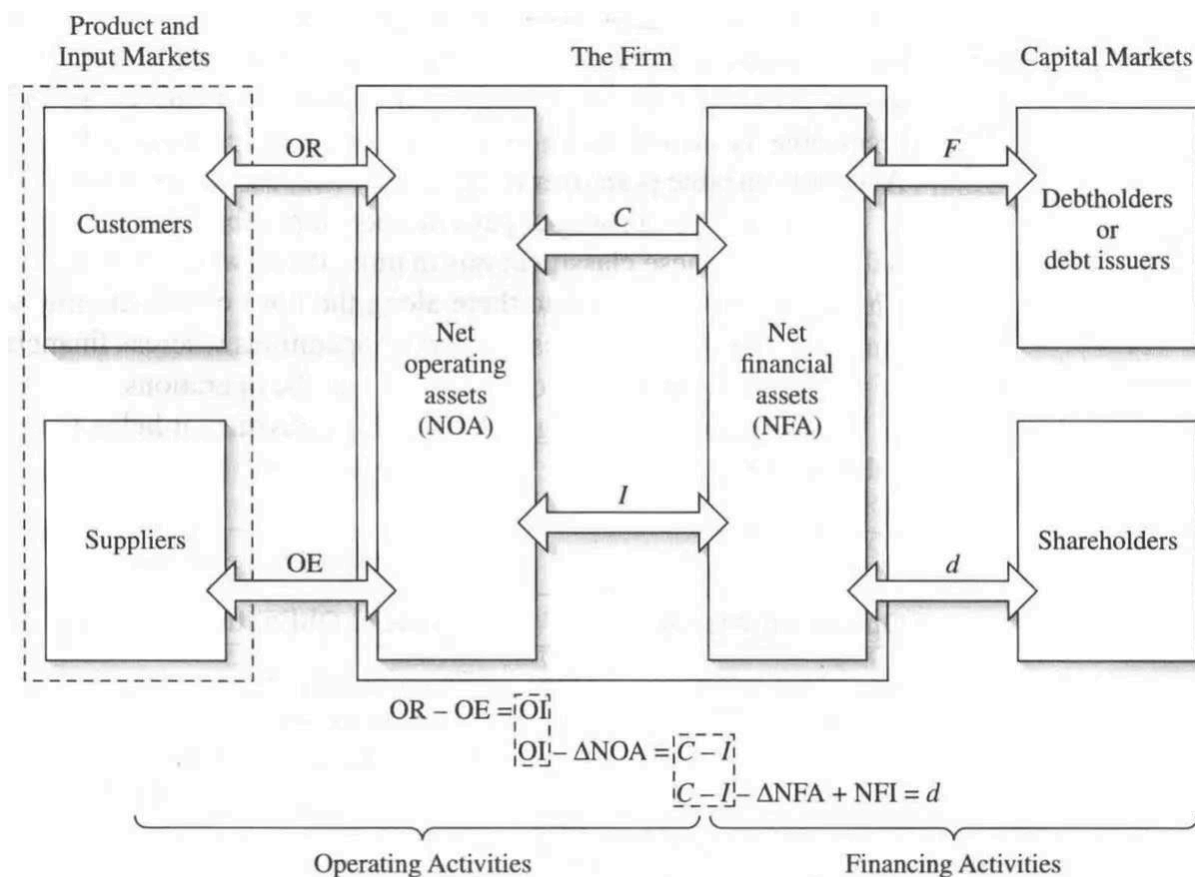
- If $C - I - i > d$, then lend or buy down own debt 借出资金/ 赎债（有钱没地方花）
- If $C - I - i < d$, then borrow or reduce lending 借入/ 出售债权性投资（拉饥荒、砸锅卖铁）

Balance Sheet Reformulated 资产负债表的重构

Balance Sheet			
Assets		Liabilities and Equity	
Operating assets	OA	Operating liabilities	OL
Financial assets	FA	Financial obligations	FO
		Common stockholders' equity	CSE
Total assets	<u>OA + FA</u>	Total claims	<u>OL + FO + CSE</u>

Reformulated Balance Sheet			
Operating Assets		Financial Obligations and Owners' Equity	
Operating assets	OA	Financial obligations	FO
Operating liabilities	(OL)	Financial assets	(FA)
		Net financial obligations	NFO
		Common shareholders' equity	CSE
Net operating assets	<u>NOA</u>		<u>NFO + CSE</u>

- Net operating assets(NOA) = $OA - OL$
- Net financial assets(NFA) = $FA - FO$
Usually $NFA < 0$, in which case it is Net financial obligations(NFO)
- Common shareholders' equity(CSE) = $NOA + NFA = NOA - NFO$
- 负债表平衡等式: $CSE = NOA - NFO$



Key:

F = Net cash flow to debtholders and issuers	NOA = Net operating assets
d = Net cash flow to shareholders	OR = Operating revenue
C = Cash flow from operations	OE = Operating expense
I = Cash investment	OI = Operating income
NFA = Net financial assets	NFI = Net financial income

The SOURCES & DISPOSITION of Free cash flow

- 来源: Free cash flow = Operating income - Δ Net operating assets

$$C - I = OI - \Delta NOA$$

$$= (OR - OE) - \Delta NOA$$

- 去向1: Free cash flow = Δ Net financial assets - Net financial income + Net dividends

$$C - I = \Delta NFA - NFI + d$$

- 去向2(承担负债): Free cash flow = Net financial expenses - Δ Net financial obligations + d

$$C - I = NFE - \Delta NFO + d$$

将 去向1&2 等式进行变化，得到股利 (股利支付净额) $d =$
Net dividends的来源

NOA和NFO的影响因素

一切均基于上述等式，下述与上式一一对应

- 净经营性资产NOA

$$\begin{aligned}NOA_t &= NOA_{t-1} + OI_t - (C_t - I_t) \\ \Delta NOA_t &= OI_t - (C_t - I_t)\end{aligned}$$

- 净金融性资产NFA

$$\begin{aligned}NFA_t &= NFA_{t-1} + NFI_t + (C_t - I_t) - d_t \\ \Delta NFA_t &= NFI_t + (C_t - I_t) - d_t\end{aligned}$$

- 净金融性负债NFO

$$\begin{aligned}NFO_t &= NFO_{t-1} + NFE_t - (C_t - I_t) + d_t \\ \Delta NFO_t &= NFE_t - (C_t - I_t) + d_t\end{aligned}$$

- 普通股股东权益CSE=Common stockholders' equity

$$\Delta CSE = \Delta NOA - \Delta NFO$$

将上式代入得：

$$\begin{aligned}\Delta CSE_t &= [OI_t - (C_t - I_t)] - [NFE_t - (C_t - I_t) + d_t] \\ &= (OI_t - NFE_t) - d_t \\ CSE_t &= CSE_{t-1} + (OI_t - NFE_t) - d_t\end{aligned}$$

where $CSE_{t-1} = CSE_t + \text{Comprehensive income} - \text{Net dividends}_t$



$$(1) \text{NOA}_{t-1}$$

$$\text{OI}_t - (C_t - I_t)$$

$$\text{NOA}_t$$

$$(2) \underline{\text{NFO}_{t-1}}$$

$$\underline{\text{NFE}_t - (C_t - I_t) + d_t}$$

$$\underline{\text{NFO}_t}$$

$$(3) \underline{\underline{\text{CSE}_{t-1}}}$$

$$\underline{\underline{\text{OI}_t - \text{NFE}_t - d_t}}$$

$$\underline{\underline{\text{CSE}_t}}$$

$$\underbrace{\hspace{1.5cm}}_{\text{Earnings}}$$

Formula of Ch9

Basic Calculations

Review:

- Net dividend = Dividends + Share repurchases – Share issues
- Comprehensive Earnings = CSE (end) – CSE (beginning) + Net dividend
- ROCE = $\frac{Earn_t}{B_{t-1}}$, where Earn = Comprehensive income (earnings), B = CSE

新一点，但又不完全新

- 普通股权益 = 所有股权益 - 优先股权益

$$\text{Common equity} = \text{total equity} - \text{preferred equity}$$

Dirty-surplus accounting 非清洁盈余会计

值将损益项目绕过利润表，直接报告在股东权益变动表中的会计处理方法。

Reformulation of the Equity Statement 权益表的重构

Step1: 使其仅反映普通股股东权益：

扣除优先股 (Preferred stock)、少数股东权益 (Noncontrolling interests)、应付股利 (Dividends payable) 金额；

Step2: 计算股利支付净额 (*net dividend*) :

这里的net dividend实际上是Cash dividends

$$\text{Cash dividends} = \text{Dividends reported} - \Delta \text{Dividends payable}$$

Step3: 计算综合收益 (*Comprehensive income*):

$$\text{同上: } CI = \Delta \text{CSE} + d$$

具体项目及算法如下：

Reformulated Statement of Common Shareholders' Equity

Beginning book value of common equity	
+ Net effect of transactions with common shareholders	
+ Capital contributions (share issues)	
– Share repurchases	
– Dividends	
= Net cash contribution (negative net dividends)	
+ Effect of operations and nonequity financing	
+ Net income (from income statement)	
+ Other comprehensive income	
– Preferred dividends	
= Comprehensive income available to common	
Closing book value of common equity	

格式如下：

Reformulated Statement of Common Equity		
Balance at May 31, 2009		\$8,814.5
Transactions with shareholders		
Stock issued for stock options	379.6	
Stock issued to employees (net)	37.1	
Stock repurchased	(754.3)	
Cash dividends	(505.5)	(843.1)
Comprehensive income		
Net income reported	1,906.7	
Net translation gain (loss)	(159.2)	
Net hedging gains (loss)	6.5	1,754.0
Stock-based compensation		159.0
Balance at May 31, 2010		<u>\$9,884.4</u>

个人理解

Transaction with shareholders <=> Net dividend

= Cash dividends + Share repurchases + Share issues

其中：Share issues = exercised (行权) + Loss

Loss on exercise before tax = Market price of shares issued in exercise - Exercise price

Tax benefit = Loss on exercise before tax * tax rate

Comprehensive income <=> 除了Net dividend以外的其他项目

中 Loss on exercise of employee options

= Loss on exercise before tax * (1 - tax rate)

= Loss on exercise before tax - Tax benefit

Formula of Ch10

Reformulated balance sheet

保证负债表平衡等式: $CSE = NOA - NFO$ 成立

The Reformulated Balance Sheet	
Assets	Liabilities and Stockholders' Equity
Financial assets:	Financial liabilities:
Cash equivalents	Short-term borrowings
Short-term investments	Current maturities of long-term debt
Short-term notes receivable (?)	Short-term notes payable (?)
Long-term debt investments	Long-term borrowing (bank loans, bonds payable, notes payable)
	Lease obligations
	Preferred stock
Operating assets:	Operating liabilities:
All else	All else
	Minority (noncontrolling) interest
	Common equity

练习题举例:

Reformulated balance sheet

Operating assets	\$547	Financial obligations	\$190
Operating liabilities	<u>132</u>	Financial assets	<u>145</u>
		Net financial obligations	45
		Common shareholders' equity	<u>370</u>
Net operating assets	<u>\$415</u>		<u>\$415</u>

Operating liabilities = \$322 - 190 = \$132 million.

Reformulated comprehensive income statement

Reformulated Comprehensive Income Statement

Net sales

– Expenses to generate sales

Operating income from sales (before tax)

– Tax on operating income from sales

+ Tax as reported

+ Tax benefit from net financial expenses

– Tax allocated to other operating income

Operating income from sales (after tax)

± Other operating income (expense) requiring tax allocation

Restructuring charges and asset impairments

Merger expenses

Gains and losses on asset sales

Gains and losses on security transactions

– Tax on other operating income

± After-tax operating items

Equity share in subsidiary income

Operating items in extraordinary income

Dirty-surplus operating items in Table 9.1

Hidden dirty-surplus operating items

Operating income (after tax)

– **Net financial expenses after tax**

+ Interest expense

– Interest revenue

± Realized gains and losses on financial assets

= Net taxable financial expense before tax

– Tax benefit from net financial expenses

= Net taxable financial expenses after tax

± Gains and losses on debt retirement

± Dirty-surplus financial items in Table 9.1 (including preferred dividends)

± Hidden dirty-surplus financing items

– Minority interest

= **Comprehensive income to common**

练习题举例：

Reformulated income statement

Revenue		\$4,356
Cost of goods sold		<u>3,487</u>
Gross margin		869
Operating expenses		<u>428</u>
Operating income		441
Net financing expense:		
Interest expense	\$132	
Interest income	<u>56</u>	<u>76</u>
Earnings		<u>\$ 365</u>

Top-down & Bottom-up methods of TAX ALLOCATION

Exempli gratia: tax rate=35%

GAAP Income Statement		Top-Down Tax Allocation		Bottom-Up Tax Allocation	
Revenue	\$4,000	Revenue	\$4,000	Net income	\$350
Operating expenses	(3,400)	Operating expenses	<u>(3,400)</u>	Interest expense	\$100
Interest expense	<u>(100)</u>	Operating income before tax	600	Tax benefit	<u>35</u>
Income before tax	500	Tax expense:		Operating income after tax	<u>\$415</u>
Income tax expense	<u>(150)</u>	Tax reported	\$150		
Net income	<u>\$ 350</u>	Tax benefit for interest	<u>35</u>		
		(\$100 x 0.35)	<u>(185)</u>		
		Operating income after tax	<u>\$ 415</u>		

1. Top-down 自上而下所得税分摊

$$\text{Tax benefit for interest} = \text{Net interest expense} \times \text{tax rate}$$

$$\text{Tax expense} = \text{Tax reported} + \text{Tax benefit for interest}$$

$$\text{Operating income after tax} = \text{Operating income before tax} - \text{Tax expense}$$

一般此时没到底，则用来检验有：

$$\text{Net income} = \text{Earnings} = \text{Operating income before tax} - \text{Net interest}$$

$$\text{Net interest} = \text{Interest expense} - \text{Tax benefit}$$

$$= \text{Interest expense} \times (1 - \text{tax rate})$$

2. Bottom-up 自下而上所得税分摊

$$\begin{aligned}\text{Net interest after tax} &= \text{Net interest expense} \times \text{tax rate} \\ \text{Operating income after tax} &= \text{Net income} + \text{Net interest after tax}\end{aligned}$$

基于上述，根据statutory tax rate (法定税率) 计算effective tax rate (真实税率)

1. on operating income (类似 Top-down)

$$\begin{aligned}\text{Effective tax rate on operating income} &= \frac{\text{Tax on operating income}}{\text{Operating income}} \\ \text{Tax on operating income} &= \text{Tax reported} + \text{Tax benefit on interest}\end{aligned}$$

2. on net income before tax

$$\begin{aligned}\text{Effective tax rate on income before tax} &= \frac{\text{Tax expense}}{\text{Net income before tax}} \\ \text{Net income before tax} &= \text{Operating income} - \text{Interest expense}\end{aligned}$$

Residual income from operations

$$\begin{aligned}\text{Residual operating income} &= \text{Operating income}_t - \\ &\quad (\text{Required return} \times \text{Net operating assets}_{t-1}) \\ ReOI_t &= OI_t - (\rho - 1)NOA_{t-1}\end{aligned}$$

Formula of Ch14

Review

$$\begin{aligned}
 V_0^E &= CSE_0 + PV(RE) \\
 &= CSE_0 + \frac{RE_1}{\rho_E} + \frac{RE_2}{\rho_E^2} + \frac{RE_3}{\rho_E^3} + \dots \\
 RE_t &= Earn_t - (\rho_E - 1)CSE_{t-1}
 \end{aligned}$$

Earnings Component	Book Value Component	Residual Earnings Measure
Operating income (OI)	Net operating assets (NOA)	Residual operating income: $OI_t - (\rho_F - 1) NOA_{t-1}$
Net financial expense (NFE)	Net financial obligations (NFO)	Residual net financial expense: $NFE_t - (\rho_D - 1) NFO_{t-1}$
Earnings	Common stockholders' equity (CSE)	Residual earnings: $Earn_t - (\rho_E - 1) CSE_{t-1}$

Note that $\rho_D - 1$ is the cost of capital for net debt (or the required return on net financial assets),

and $\rho_F - 1$ is the cost of capital for operating activities.

Then

$$RE_t = Earn_t - (\rho_E - 1)B_{t-1} = [ROCE_t - (\rho_E - 1)]CSE_{t-1}$$

where $ROCE_t = \frac{Earn_t}{B_{t-1}} = \frac{Earn_t}{CSE_{t-1}}$

$$ReOI_t = OI_t - (\rho_F - 1)NOA_{t-1} = [RNOA_t - (\rho_F - 1)]NOA_{t-1}$$

where $RNOA_t = \frac{OI_t}{NOA_{t-1}}$

Abnormal operating income growth (AOIG)

$$\begin{aligned}
 AOIG_t &= \text{Cum-d operating income}_t - \text{Normal operating income}_t \\
 &= [OI_t + (\rho_F - 1)(C - I)_{t-1}] - \rho_F \times OI_{t-1}
 \end{aligned}$$

where $C - I$ is free cash flow, from Ch8, $C - I = OI - \Delta NOA$

the same like AEG:

$$AOIG_t = (G_t - \rho_F) \times OI_{t-1}$$

where function G_t is cum-dividend operating income growth rate.

The short hand method:

$$AOIG_t = \Delta ReOI_t = ReOI_t - ReOI_{t-1}$$

Weight-average of cost of equity & cost of net debt (WACC)

Cost of capital for operations = WACC

$$= \left(\frac{\text{Value of equity}}{\text{Value of operations}} \times \text{Equity cost of capital} \right) + \left(\frac{\text{Value of debt}}{\text{Value of operations}} \times \text{Debt cost of capital} \right)$$

$$\rho_F = \frac{V_0^E}{V_0^F} \cdot \rho_E + \frac{V_0^D}{V_0^F} \cdot \rho_D$$

where:

$$\rho_D = \text{净负债成本} \times (1 - \text{tax rate})$$

$$\rho_E = \rho_F + \frac{V_0^D}{V_0^E} (\rho_F - \rho_D) = \rho_F + \text{Equity beta} \times \text{Market risk premium}$$

$$V_0^E + V_0^D = V_0^F$$

Formula of Ch15

Review

Valuation & Forecast

$$V_0^{NOA}(= V_0^F) = NOA_0 + \frac{ReOI_1}{\rho_F - g}$$

$$ReOI_1 = OI_1 - (\rho_F - 1)NOA_0$$

where g is the growth of valuation. If there is no-growth valuation, $g = 1$ simply.

$$\text{Forward enterprise P/E} = \frac{V_0^{NOA}}{OI_1}$$

If there is no-growth valuation, Forward enterprise P/E = $1/(\rho_F - 1)$ simply.

Enterprise P/B Ratio

(Core) OI_1 = Simple growth forecast of $OI = NOA_0 \times$ (Core) $RNOA_0$

$$V_0^E = CSE_0 + \frac{ReOI_1}{\rho_F - g}$$

$$(\text{Enterprise value}) = V_0^{NOA} = V_0^E - NFA_0 = V_0^E + NFO_0$$

$$\text{Enterprise } P/B = \frac{V_0^{NOA}}{NOA_0}$$

or

$$\text{Enterprise } P/B = \frac{RNOA_0 - (g - 1)}{\rho_F - g}$$

Therefore

$$V_0^{NOA} = \frac{RNOA_0 - (g - 1)}{\rho_F - g} \times NOA_0$$

About AOIG

$$\text{Cum-div } OI \text{ growth rate}_0 = \frac{\text{Cum-div } OI_0}{OI_{-1}} - 1$$

老太太裹脚布

$$\begin{aligned} ReOI_1 &= OI_1 - (\rho_F - 1)NOA_0 \\ &= [RNOA_1 - (\rho_F - 1)]NOA_0 \end{aligned}$$

where $OI_1 = RNOA_1 \cdot NOA_0$

$$\begin{aligned} NFE_0 &= (\rho_D - 1)NFO_0 \\ \text{Comprehensive earning} &= CI_1 = OI_1 - NFE_0 \end{aligned}$$

$$RE_1 = Earn_1 - (\rho_E - 1)CSE_0$$

2-stage growth model

$$V_0^{NOA} = OI_1 \times \frac{1}{\rho_F - 1} \left[\frac{G_2 - G_{long}}{\rho_F - G_{long}} \right]$$

where G_2 is (1 plus) the growth rate forecasted for cum-div OI 2 years ahead

G_{long} is the long-term one, which is simply equal to the GDP growth rate

Then

$$V_0^E = V_0^{NOA} - NFO$$

shabizouj

Simple Forecast	Simple Valuation of the Equity	Simple Valuation of the Operations
No-growth	$V_0^E = CSE_0 + \frac{\text{Core ReOI}_0}{\rho_F - 1}$	$V_0^{NOA} = NOA_0 + \frac{\text{Core ReOI}_0}{\rho_F - 1}$ $= \frac{OI_1}{\rho_F - 1}$
Growth	$V_0^E = CSE_0 + \frac{[\text{Core RNOA}_0 - (\rho_F - 1)]NOA_0}{\rho_F - g}$	$V_0^{NOA} = NOA_0 + \frac{[\text{Core RNOA}_0 - (\rho_F - 1)]NOA_0}{\rho_F - g}$ $= NOA_0 \times \frac{\text{Core RNOA}_0 - (g - 1)}{\rho_F - g}$ $V_0^{NOA} = OI_1 \times \frac{1}{\rho_F - 1} \left[1 + \frac{G_2 - \rho_F}{\rho_F - g} \right]$

Formula of Ch16

Residual operating income & its drivers

$$RNOA = (\text{Core sales PM} \times ATO + \frac{\text{Core other OI}}{NOA} + \frac{\text{Unusual items}}{NOA})$$

where PM is the profit margin.

$$ReOI = \text{Sales} \times [\text{Core sales PM} - \frac{\rho_F - 1}{ATO}] + \text{Core other OI} + \text{Unusual items}$$

Remind

$$ReOI_1 = OI_1 - (\rho_F - 1)NOA_0$$

where $OI_0 = Sales_0 \times PM$, $NOA_0 = Sales_1 / ATO$.

Pure equity (no debt)

$$\text{Free cash flow (FCF)} = C - I = d$$

where $F = 0$.

$$NOA = CSE (+NFO)$$

where $NFO = 0$.